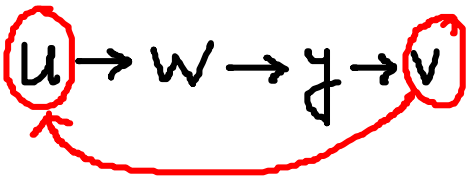
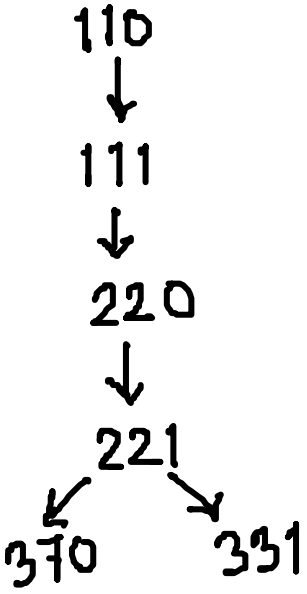
Topological Sorting

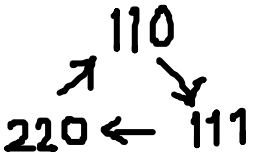


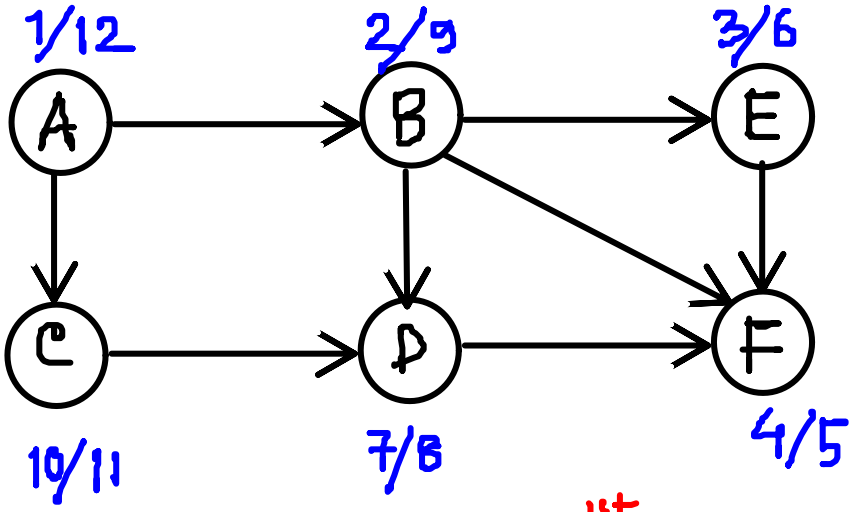


A topological sort of a graph is a linear ordering of all its vertices such that if G contains an edge (u, v), then u appears before v in the ordering.

Topological sort is the sorting of the vertices according to the departure time indescending order (largest ending time to the smallest ending time) . Through topological sort, we can vertices in such a way that all every edge goes from left to right. Topological sort can be implemented only Directed Acyclic Graph(DAG). It means the graph has to be directed and there cannot be any cycle.

Why Acyclic? →





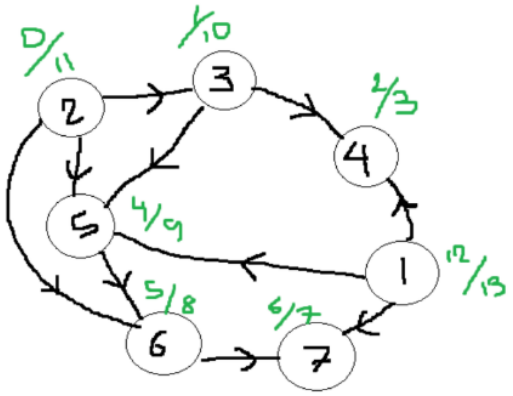
1st

A C B D E F → Topological Ordering

12 11 9 8 6 5

A → C B → D E → F → Ending time – Descending Order

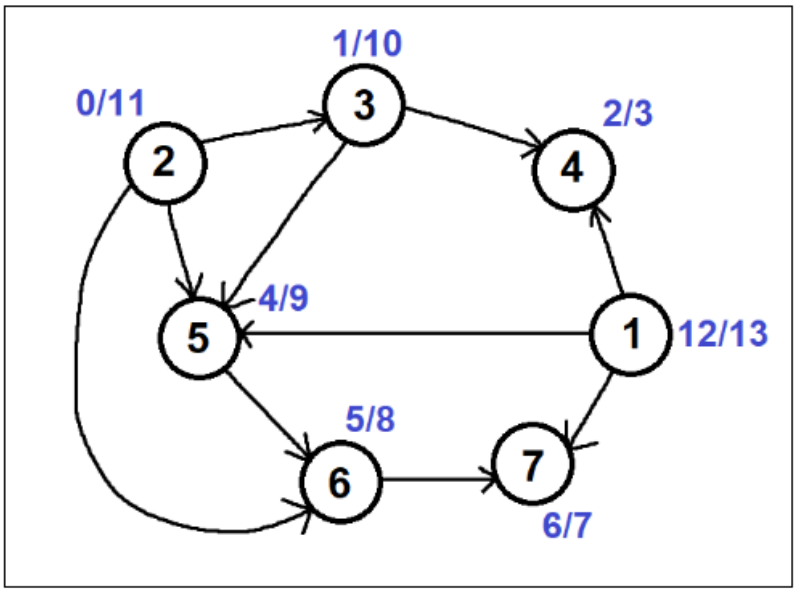
The graph below is an DAG with starting and ending times.



If the ending times are arranged in decreasing manner we will get the ordering of the vertices as shown below.

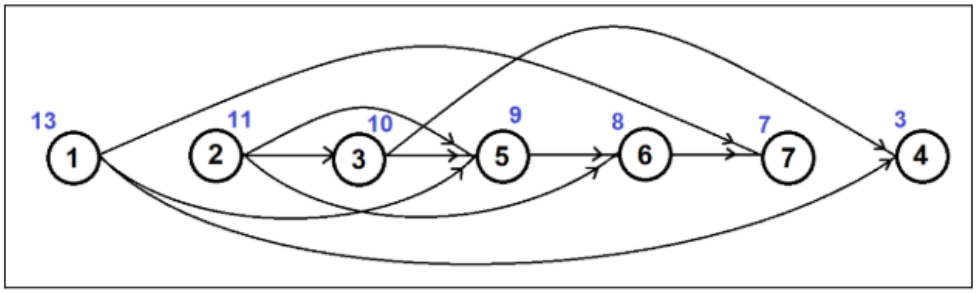
Ending Time	13	11	10	9	8	7	3
Corresponding Vertices	1	2	3	5	6	7	4

A directed graph can have multiple topological sorts.
The time complexity is O(V + E) which is linear.



If the ending times are arranged in a decreasing manner we will get the ordering of the vertices as shown below.

End Time	13	11	10	9	8	7	3
Vertices	1	2	3	5	6	7	4



Time Complexity: O(V+E) as it is same as DFS